Genetic coupling of floral rewards and signals underlies pollinator attraction

Footnotes6These authors contributed equally

Highlights

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YAB5 and CRC define nectary identity and inhibit alternative cell fates

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YAB5 activates DPE2-dependent starch remobilization for nectar secretion

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YAB5 promotes petal growth, flower opening, and MYB111-dependent UV guides

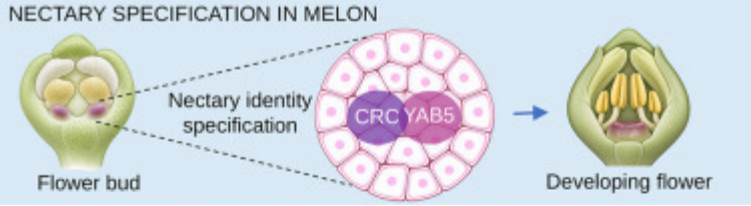
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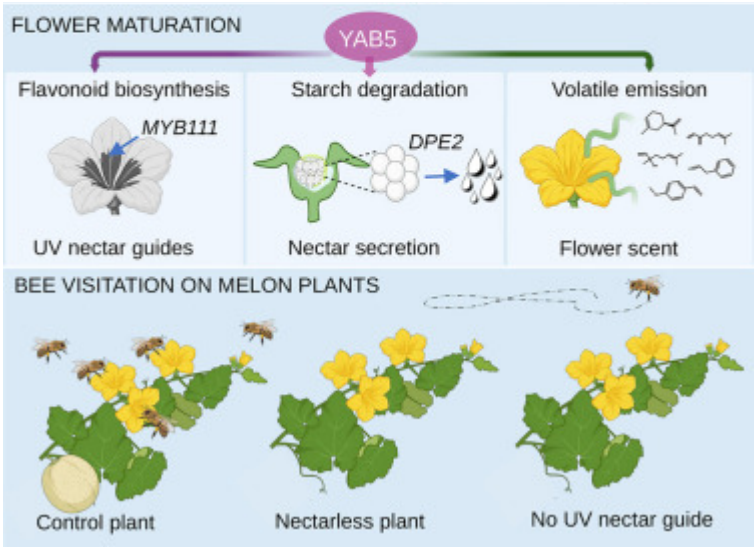
YAB5 coordinates floral reward and advertisement to promote bee visitation

Summary

Animal pollinators underpin biodiversity and global agriculture, yet the molecular mechanisms that coordinate floral rewards and signals to attract pollinators remain poorly understood. Through phenotyping melon genetic diversity, we identified *nectarless1* (*nls1*), encoding the YABBY-family transcription factor YABBY5 (YAB5), as a central integrator of pollinator-attracting traits. YAB5 interacts with CRABS CLAW (CRC) during nectary initiation to specify nectary identity by repressing alternative floral organ fates. As flowers mature, YAB5 directly activates *4-α-glucanotransferase 2* (*DPE2*) to promote starch remobilization required for nectar secretion and promotes petal growth and flower opening. In petals, YAB5 directly engages an *MYB111*-dependent flavonoid pathway that generates UV nectar guides, while indirectly influencing floral volatile emission. Disrupting *YAB5*, *DPE2*, or *MYB111* reduces pollinator visitation. Together, these findings reveal how a developmental regulator couples organ specification with organ-specific metabolic programs to synchronize floral rewards and signals at anthesis.

Graphical abstract





Keywords

nectary development

nectar

pollinators

flower maturation

cucurbits

bees

UV nectar-guide

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Rapid plant evolution driven by the interaction of pollination and herbivory

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